

LINEAR REGRESSION-Project for AIML BATCH

Project Title: Predict New York City Airbnb Prices

The Dataset

Have your students download the classic New York City Airbnb Open Data from Kaggle.

- **Search on Kaggle:** "New York City Airbnb Open Data" by AB NYC
- **What it contains:** Around 49,000 listings with a mix of numerical data (*price, minimum nights, latitude/longitude*) and categorical data (*boroughs, room types*).

Step-by-Step Student Instructions

Step 1: Exploratory Data Analysis (EDA) & Data Auditing

Before writing any ML algorithms, students must understand what they are working with.

- **Task 1.1:** Load the dataset into a Pandas DataFrame and inspect its shape, columns, and data types.
- **Task 1.2:** Identify which columns are numerical (discrete/continuous) and which are categorical (nominal/ordinal).
- **Task 1.3:** Check for missing values across all columns. Write down which columns have missing data and calculate their percentages.

Step 2: Data Cleaning & Handling Missing Values

Students must make executive data science decisions based on their EDA.

- **Task 2.1:** Look at the **last_review** column. Since it's a date string with over 10,000 missing values, decide whether to drop the column entirely or engineer it.
- **Task 2.2:** Look at the **reviews_per_month** column. If it contains missing values, investigate why they are missing (*Hint: look at the number of reviews for those rows*) and fill them with a logical replacement value.
- **Task 2.3:** Drop irrelevant identifier columns that won't help a mathematical model (e.g., URLs, host names, or unique IDs).



Step 3: Outlier Identification & Treatment

Linear Regression is highly sensitive to extreme values. Students must protect their models.

- **Task 3.1:** Create boxplots or histograms for **price** and **minimum_nights** to visually detect outliers.
- **Task 3.2:** Apply a capping or trimming strategy. Use the 95th percentile rule to remove extreme luxury penthouses (e.g., \$10,000/night listings) and long-term rental rules (e.g., minimum stays of 365+ nights) so the model focuses on the typical consumer market.

Step 4: Feature Selection & Categorical Encoding

A computer cannot run a linear regression equation on text strings like "Manhattan" or "Private room".

- **Task 4.1: Address High Cardinality.** The dataset has a **neighbourhood** column with 200+ unique values and a **neighbourhood_group** column with 5 values. Explain why keeping the 200+ unique values will break the model's simplicity (*The Curse of Dimensionality*), and drop the hyper-specific one.
- **Task 4.2:** Convert the remaining categorical columns (**neighbourhood_group** and **room_type**) into numerical format using One-Hot Encoding. Ensure they drop the first column to avoid the "*Dummy Variable Trap*."

Step 5: Feature-Target Separation & Validation Split

Preparing the data for the "Final Exam."

- **Task 5.1:** Separate the cleaned data into features (\$X\$) and the target variable (\$y\$).
- **Task 5.2:** Split the data into a Training Set (80%) and a Testing Set (20%). Ensure they use a specific reproducibility seed (**random_state**) so their results can be verified.

Step 6: Model Training & Evaluation



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- **Task 6.1:** Import, initialize, and fit a standard Linear Regression model using the training data.
- **Task 6.2:** Use the trained model to generate price predictions on the hidden testing data.
- **Task 6.3:** Calculate three key metrics to grade the model:
 - **R-Squared (R^2) Score:** To find out what percentage of the price variance the model actually understands.
 - **Mean Absolute Error (MAE):** To find the average mistake the model makes in actual US dollars.
 - **Mean Squared Error (MSE) / RMSE:** To analyze how heavily the model is being punished for large mistakes.

Step 7: Residual Analysis & Interpretation

- **Task 7.1:** Plot a scatter plot of *Actual Prices* vs. *Predicted Prices*. Draw a $y=x$ diagonal line to show what a "perfect" prediction would look like.
- **Task 7.2:** Plot a *Residual Plot* (Errors vs. Predicted values) or an error distribution histogram.
- **Task 7.3:** Answer the final project question: *Does the model perform better on cheap listings or expensive listings, and why?*

Expected Student Deliverables

1. **A Clean Jupyter / Google Colab Notebook:** Well-commented code blocks following these exact steps.
2. **A 1-Page Summary Report:** Students must state their final R^2 score and MAE in plain English, explaining exactly what those numbers mean to a non-technical business stakeholder.

